

資料結構 Homework 2

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1.

$n=4$

0.1.2.3 1.0.2.3 2.1.0.3 2.3.1.0
 0.1.3.2 1.0.3.2 2.1.3.0 3.2.1.0
 0.2.1.3 1.2.0.3
 0.2.3.1 1.2.3.0
 0.3.2.1 1.3.2.0

2.

top	Q[0]	Q[1]	Q[2]	Q[3]	Comments
-1					stack is empty
0	J1				push J1, top = 0
1	J1	J2			push J2, top = 1
2	J1	J2	J3		push J3, top = 2
1	J1	J2			pop Q[top], top-1
2	J1	J2	J4		push J4, top = 2
3	J1	J2	J4	J5	push J5, top = 3
3	J1	J2	J4	J5	stack is full (top ≥ 3)

3.

front	rear	Q[0]	Q[1]	Q[2]	Q[3]	Comments
-1	-1					queue is empty
-1	0	J1				push J1 = queue[++rear]
-1	1	J1	J2			push J2 = queue[++rear]
-1	2	J1	J2	J3		push J3 = queue[++rear]
0	2		J2	J3		J1 被刪除
0	3		J2	J3	J4	push J4 = queue[++rear]
0	3		J2	J3	J4	queue is full.
0	3		J2	J3	J4	queue is full.

4.

front	rear	Q[0]	Q[1]	Q[2]	Q[3]	Comments
0	0					queue is empty
0	1		J1			add J1 , rear++
0	2		J1	J2		add J2 , rear++
0	3		J1	J2	J3	add J3 , rear++
1	3			J2	J3	delete ^{J1} , front++
1	0	J4		J2	J3	add J4 , rear++
1	0	J4		J2	J3	circular queue is full.
2	0	J4			J3	delete ^{J2} , front++

3 0 J4

delete J3

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5.

(a)

$$\begin{aligned}
 & (A+B) * C + D / (E + F * G) - H \\
 &= (((A+B) * C) + (D / (E + (F * G)))) - H \\
 &= AB + C * DEFG * + / + H - *
 \end{aligned}$$

(b)

$$\begin{aligned}
 & ((A \& B) \parallel C) \parallel ((E > F)) \\
 &= AB \& C \parallel EF > ! \parallel *
 \end{aligned}$$

6.

$$ABC < CD > \parallel ! \& \& ! C E < \parallel$$

$$\Rightarrow ((!(A \& \& !(B < C) \parallel (C > D))) \parallel (C < E))$$